



BINARY SEARCH 2

"Motivation is what gets you started. Habit is what keeps you going."

~ Jim Ryun



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Good

Morning

Today's content

01. $\text{Sqrt}(N)$
02. Search in sorted Rotated Array
03. A* magical no.
04. Median in a matrix

01. Given a positive no. N , find $\text{sqrt}(N)$.

Given N , find greater i such that $i * i \leq N$.

$\text{floor}(\text{sqrt}(n))$

$$\text{sqrt}(25) \Rightarrow 5$$

$$\text{sqrt}(16) \Rightarrow 4$$

$$\text{sqrt}(52) \Rightarrow 7$$

$$\text{sqrt}(48) \Rightarrow 6$$

Idea 1 = Start iterating from $i=1$ & keep on updating your ans until $i * i \leq n$

$$\underline{\underline{n = 40}}$$

<u>i</u>	$i * i \leq n$	<u>ans</u>
1	$1 * 1 \leq 40$	1
2	$2 * 2 \leq 40$	2
3	$3 * 3 \leq 40$	3
4	$4 * 4 \leq 40$	4
5	$5 * 5 \leq 40$	5
6	$6 * 6 \leq 40$	6
7	$7 * 7 \leq 40$	→ Break

Ans = 6

01. Target $\rightarrow \text{sqrt}(n)$

02. search $\rightarrow$ 1 N
space lo hi

03. How to reduce search space

Case I



if ($\text{mid} * \text{mid} == n$)
return mid;

Case II



if ($m * m < n$)
| ans = m;
| // go to right
3

Case III



if ($m * m > n$)
| // go to left
3

$N = 50$

lo hi mid Compare $\text{mid} * \text{mid}$ & N

1 50 25

$25 * 25 > 50$: $hi = m - 1$

1 24 12

$12 * 12 > 50$: $hi = m - 1$

1 11 6

$6 * 6 < 50$ **ans = 6**

$lo = m + 1$

7 11 9

$9 * 9 > 50$: $hi = m - 1$

7 8 7

$7 * 7 < 50$ **ans = 7**

$lo = m + 1$

8 8 8

$8 * 8 > 50$: $hi = m - 1$

8 7 $\longrightarrow$ stop

Ans = 7

$lo = 1$ $hi = N$ $ans = 0$

while ($lo \leq hi$) {

 int $m = (lo + hi) / 2$

 if ($m * m == n$) {

 return m;

 else if ($m * m < n$) {

$ans = m$;

$lo = m + 1$;

 else {

$hi = m - 1$;

Tc: $O(\log N)$

Sc: $O(1)$

 }
return ans;

* sorted arr $[] = \{ 3, 9, 14, 16, 20, 23, 35 \}$

Given K , can K be present inside this arr $[]$ or not?

$K = 16 \longrightarrow \underline{\text{Yes}}$

$K = 60 \longrightarrow \text{No}$

Claim $\rightarrow$ If I have a sorted arr, then I can decide whether K will be present or not.

* Rotated sorted arr

arr $[] = \{ 1, 2, 3, 4, 5 \}$

$K = 0 = \{ 1, 2, 3, 4, 5 \}$

$K = 1 = \{ 5, 1, 2, 3, 4 \}$

$K = 2 = \{ 4, 5, 1, 2, 3 \}$

$K = 3 = \{ 3, 4, 5, 1, 2 \}$

$K = 4 = \{ 2, 3, 4, 5, 1 \}$

Claim $\rightarrow$ If we are rotating a sorted arr, then i'll always get atleast one half sorted for any particular K

* Given an input $ar[]$, formed by rotating a distinct sorted array right to left.

Search ele & return index if it is present
else return -1 .

$ar[] =$

0	1	2	3	4	5	6	7	8	9	10	11	12	13
10	11	12	13	17	20	23	25	26	1	3	5	6	8

$K = 17 \longrightarrow \underline{Ans = 4}$

Idea $\rightarrow$ Linear search & check $A[i] == K$

Tc: $O(n)$

Sc: $O(1)$

01. Target $\Rightarrow K$

02. Search space = 0 to $n-1$
 lo hi

$ar[] =$

0	1	2	3	4	5	6	7	8	9	10	11	12	13
10	11	12	13	17	20	23	25	26	1	3	5	6	8

$\uparrow$ $\uparrow$ $\uparrow$
 lo m hi

lo

hi

mid

0

13

6

: left half is sorted

$hi: m-1$

0

5

2

: left half is sorted

K is not present ($l \dots m$)

$lo = m+1$

3 5 4 $\longrightarrow$ return ans = 4

ar [] =

0	1	2	3	4	5	6	7	8	9	10	11	12	13
10	11	12	13	17	20	23	25	26	1	3	5	6	8

K = 26

$\uparrow$ $\uparrow$ $\uparrow$
lo m hi

<u>lo</u>	<u>hi</u>	<u>m</u>
0	13	6

: Left half is sorted

lo = m + 1

7	13	10
---	----	----

: Right half is sorted

hi = m - 1

7	9	8
---	---	---

$\longrightarrow$ Ans = 8

ar [] =

0	1	2	3	4	5	6	7	8	9	10	11	12	13
10	11	12	13	17	20	23	25	26	1	3	5	6	8

K = 8

lo	hi	mid
0	13	6

: Left half is sorted

lo = m + 1

7	13	10
---	----	----

: Right half is sorted

lo = m + 1

11	13	12
----	----	----

: Left half is sorted

lo = m + 1

13	13	13
----	----	----

$\longrightarrow$ return Ans = 13

lo = 0 hi = n - 1

while (lo ≤ hi)

int m = (lo + hi) / 2

if (A[m] == tar) return m;

else if (A[lo] ≤ A[m]) { lo...m is sorted }

if (A[lo] ≤ tar && tar < A[m]) hi = m - 1

else lo = m + 1;

else { m...hi is sorted }

if (A[m] < tar && tar ≤ A[hi]) lo = m + 1

else hi = m - 1;

return -1;

8:09 AM → 8:19 AM

Q3 A* Magical no.

A number is magical if it is divisible by B or C

B = 2
C = 3
A = 8 } Find 8th magical no. for 2 & 3.

1 2 3 4 5 6 7 8 9 10 11 12
Ans = 12

$B = 3$
 $C = 1$
 $A = 4$

1 2 3 4 5 6 7 8 9
 $C = 1$ $C = 2$ $C = 3$ $C = 4$
Ans = 9

$B = 3$
 $C = 17$
 $A = 2$

1 2 3 4 5 6
Ans = 6

Brute force → Consider all numbers from 1 & check if it is divisible by B or C.

→ If it is → $count = count + 1$;

→ $count == A$ } return ans

No. of iterations = $A * \min(B, C)$

Max possible no. = $A * (\min B, C)$

Search space = $\left[1 \quad A * \min(B, C) \right]$
 $\min(B, C)$

Target = $A * \text{magical no.}$

Q $B=5$, $C=3$, find the count of magical no from 1 to 35

3, 5, 6, 9, 10, 12, 15, 18, 20, 21, 24, 25, 27, 30, 33, 35

$$\text{Mul of 3 in range } [1 \rightarrow 35] = \frac{35}{3} \approx 11$$

$$\text{Mul of 5 in range } [1 \rightarrow 35] = \frac{35}{5} = 7$$

$$\text{Multiples of 3 \& 5 in range } [1 \rightarrow 35] = \frac{35}{15} = 2$$

$$\text{Ans} = 11 + 7 - 2 = 16$$

Q count the magical no. $[1 \rightarrow 100]$ $B=9$ $C=12$

$$= \frac{100}{9} + \frac{100}{12} - \frac{100}{\text{LCM}(9,12)}$$

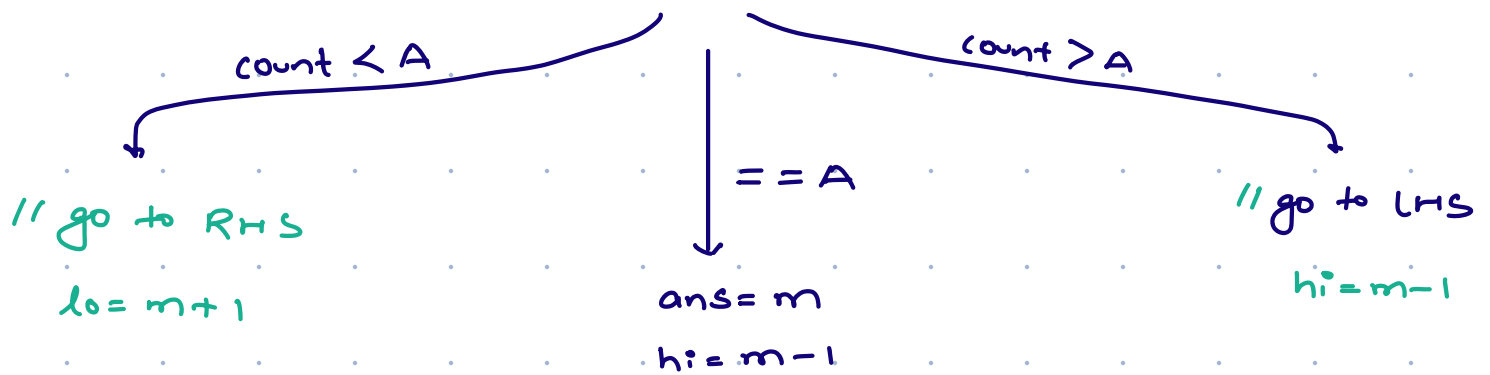
$$= \frac{100}{9} + \frac{100}{12} - \frac{100}{36} = 11 + 8 - 2 = \underline{\underline{17}}$$

$$\text{LCM}(a,b) * \text{GCD}(a,b) = a * b$$

Back to Original question

m

For this m value,
 $\text{count magical} \leq m$



Q $B=5, C=7$ $A=3$

1 2 3 4 5 6 7 8 9 10 11 12 13 14

Range = $[1 \text{ to } 15]$

<u>lo</u>	<u>hi</u>	<u>mid</u>
1	15	8
9	15	12

$\text{count magical no.} \leq \text{mid}$

$$\frac{8}{5} + \frac{8}{7} - \frac{8}{35} = 2$$

$lo = m + 1$

$$\frac{12}{5} + \frac{12}{7} - \frac{12}{35} = 3$$

$ans = 12$

$hi = m - 1$

9 11 10

$$\frac{10}{5} + \frac{10}{7} - \frac{10}{35} = 3$$

ans = 10

9 9 9

hi = m - 1

$$\frac{9}{5} + \frac{9}{7} - \frac{9}{35} = 2$$

lo = m + 1

10 9 $\longrightarrow$ stop

lo = 1

hi = A * min(B, C)

int LCM = getLCM(B, C)

while (lo <= hi)

mid = (lo + hi) / 2

int count = countMagical(mid, B, C, LCM)

if (count < A)

lo = m + 1;

else if (count > A)

hi = m - 1

else {

ans = m;

hi = m - 1;

Tc: $O(\log(\text{searchspace}) + \text{LCM})$

Tc: $O(\log(A * \min(B, C)) + \log(\min(B, C)))$

Sc: $O(1)$

* Median of an array

→ Middle element of sorted array

$$A[] = \{1, 4, 5\}$$

$$B[] = \{2, 3\}$$

Ans = 3

$$ar[] = \{1, 2, 3, 4, 5\}$$

$$A[] = \{4\}$$

$$B[] = \{1, 3, 2\}$$

Ans = 2.5

$$ar[] = \{1, 2, 3, 4\}$$

$$\frac{2+3}{2} = 2.5$$

Given two sorted arrays, figure out the median

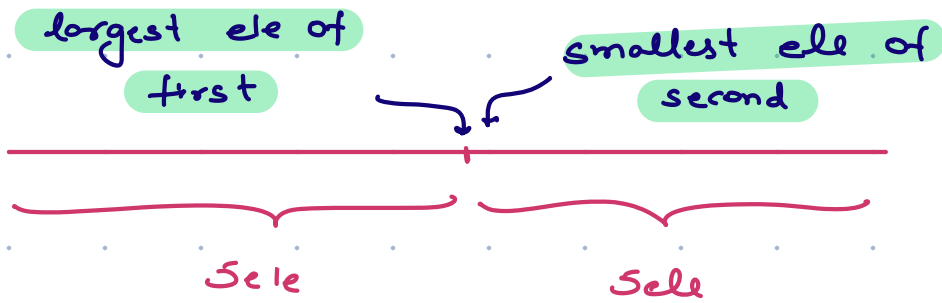
Idea 1 → Merge 2 sorted arrays & return the middle element

TC: $O(n+m)$

$$A[] = \{1, 3, 4, 7, 10, 12\}$$

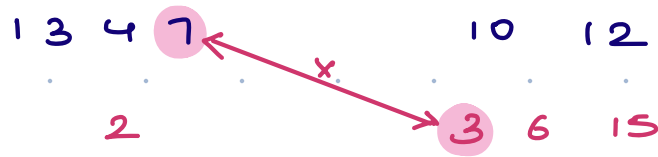
$$B[] = \{2, 3, 6, 15\}$$

Size = 10

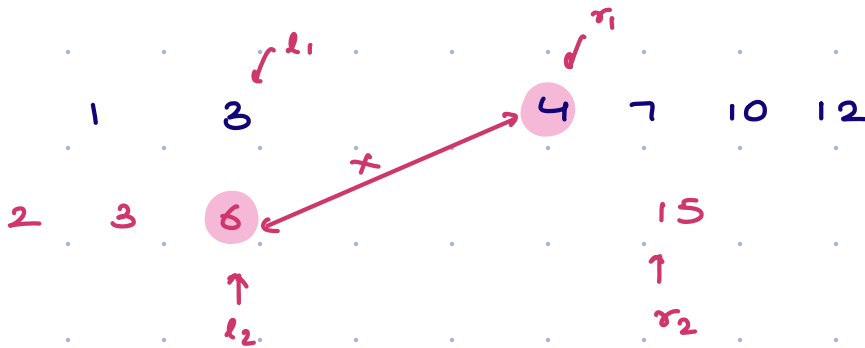


$$\text{median} = \frac{(l+s)}{2}$$

Q1. Scenario 1 = 4 ele from 1st arr



Q2. Scenario 2 = 2 ele from 1st arr



$$l_1 \leq r_1$$

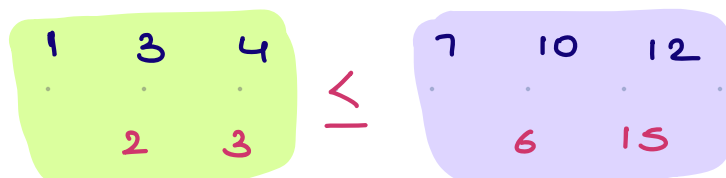
$$l_2 \leq r_2$$

$$l_1 \leq r_2$$

$$l_2 \leq r_1$$

} correct split

Q3. Scenario 3 = Pick 3 ele from 1st arr



$$\text{Ans} = \frac{4+6}{2} = 5$$

Idea $\rightarrow$ Apply binary search on array A & figure out how many ele do we have to take from A array on LHS

$A[] = \{ \overset{l_1}{7}, \overset{r_1}{12}, 14, 15 \}$ $\leftarrow$ Applying BS on A
_{0 1 2 3}

$B[] = \{ 1, 2, 3, 4, \overset{l_2}{9}, \overset{r_2}{11} \}$

Total no. of ele = 10

no. of ele in = 5
 one side

search space = [0 4]

lo	hi	mid	$\longrightarrow$	
0	4	2		choose 2 ele from A & 3 ele from B

$l_1 \leq r_2$ X : go to LHS

$hi = mid - 1$

0	1	0	$\longrightarrow$	choose 0 ele from A & 5 ele from B
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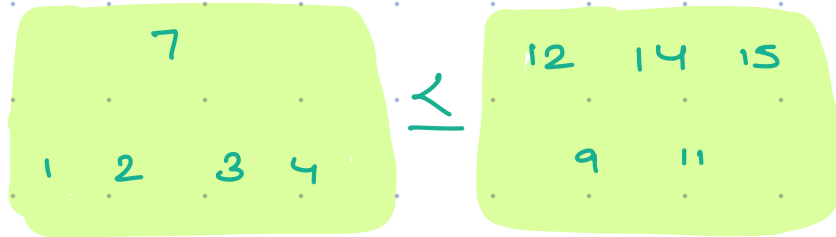
$l_2 \leq r_1$ X : go to RHS

: $lo = mid + 1$

1 1 1 $\longrightarrow$ choose 1 ele from A &

4 ele from B.

valid distribution



$$\text{ans} = \frac{\max(l_1, l_2) + \min(r_1, r_2)}{2}$$

$$l_0 = 0$$

$$h_i = \min(\text{len of } A, \frac{n+m}{2})$$

} A is smaller array

while ($l_0 \leq h_i$)

$$\text{int cut1} = (l_0 + h_i) / 2$$

$$Tc: O(\log(A.size()))$$

$$\text{int cut2} = \left(\frac{n+m+1}{2} \right) - \text{cut1}$$

$$Sc: O(1)$$

$$l_1 = (\text{cut1} == 0) ? -\infty : A[\text{cut1} - 1]$$

$$l_2 = (\text{cut2} == 0) ? -\infty : B[\text{cut2} - 1];$$

$\tau_1 = (\text{cut}_1 == n) ? \infty : A[\text{cut}_1]$

$\tau_2 = (\text{cut}_2 == m) ? \infty : B[\text{cut}_2]$

if ($l_1 \leq \tau_2$ && $l_2 \leq \tau_1$)

if ($n+m \% 2 == 0$)

return $\frac{\max(l_1, l_2) + \min(\tau_1, \tau_2)}{2}$

3

else

return $\max(l_1, l_2)$

2

else if ($l_1 > \tau_2$) { $hi = m - 1$ } : // going for less no. of elements in A

else {

2

$lo = m + 1$

// going for more elements from A array.